

GRADE 9 MATHEMATICS · ALGEBRA

Polynomials

THE BIG IDEA

Only **like terms** combine —
exponent. Everything else s

SUBTRACTING POLYNOMIALS

Distribute the negative sign to **every** term in the

$$(a - b) - (c - d) = a - b -$$

**– same variable, same
stays separate.**

second bracket, then collect like terms.

$$c + d$$

WORKED EXAMPLE

Find: $(4x^2 + 3x - 1) - (2x^2 - x + 6)$

Distribute the negative: $4x^2 + 3x - 1 - 2x^2 + x - 6$

Collect: $(4-2)x^2 + (3+1)x + (-1-6)$

Result: $2x^2 + 4x - 7$

⚠ WATCH OUT FOR

MISTAKE

$$(x + 2)(x + 3) = x^2 + 6$$

MISTAKE

$$x \times x = 2x$$

$$x^2 + x - 6$$

FIX

Needs 4 partial products: $x^2 + 5x + 6$

FIX

$$x \times x = x^2$$



SELF-CHECK

Can you expand $(2x - 3)(x + 5)$ using the area from $(6x^2 + 3x - 1)$?

a model? Can you subtract $(4x^2 - x + 2)$